

MUMBLES HEAD, United KingdomWMO: **036090**Lat: **51.5656N**Lon: **3.9817W**Elev: **141**StdP: **14.62**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.7	33.6	22.9	17.4	34.8	25.5	19.6	36.5	61.4	47.5	55.3	47.6	14.1	30	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	72.8	64.1	70.2	62.8	68.0	61.8	65.3	70.0	64.1	68.0	63.0	66.4	10.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	88.7	66.6	62.7	85.6	65.5	61.7	82.5	64.8	30.2	70.3	29.4	68.2	28.6	66.3	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 48.7	40.5	34.1	DB	29.6	77.7	2.9	3.3	27.5	80.0	25.8	81.9	24.2	83.8	22.1	86.2	(4)
(5)			WB	27.0	67.9	2.8	1.9	25.0	69.2	23.4	70.3	21.9	71.4	19.9	72.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	43.8	43.1	45.2	49.2	53.9	58.9	62.4	62.2	59.3	55.0	49.2	45.0	(6)	
	DBStd	8.01	4.29	4.22	3.84	3.98	3.80	3.70	3.58	3.09	3.19	3.98	4.51	4.94	(7)	
	HDD50	843	193	195	153	59	7	0	0	0	0	10	65	161	(8)	
	HDD65	4666	658	614	614	474	346	189	99	97	173	310	473	619	(9)	
	CDD50	1688	0	0	4	35	126	266	385	380	279	164	42	7	(10)	
	CDD65	37	0	0	0	0	5	18	12	2	0	0	0	0	(11)	
	CDH74	55	0	0	0	0	0	10	25	20	0	0	0	0	(12)	
	CDH80	4	0	0	0	0	0	1	1	2	0	0	0	0	(13)	
(14) Wind	WSAvg	15.6	17.7	16.3	15.1	14.0	14.7	14.1	13.8	14.6	14.9	17.4	16.6	17.6	(14)	
(15) Precipitation	PrecAvg	47.2	5.2	3.4	3.6	2.8	2.8	3.0	2.8	3.9	4.2	5.0	5.5	5.3	(15)	
	PrecMax	60.7	10.0	8.0	8.9	6.7	5.9	8.5	8.0	9.5	9.4	11.1	11.8	12.8	(16)	
	PrecMin	33.9	0.3	0.0	0.5	0.2	0.2	0.4	0.4	0.3	0.9	0.7	2.5	1.3	(17)	
	PrecStd	6.5	2.4	1.9	2.0	1.5	1.5	1.8	1.6	2.2	1.9	2.4	2.1	2.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	52.6	57.1	64.2	69.4	74.7	77.1	76.5	70.6	65.0	58.7	55.4	(19)	
		MCWB	50.5	48.5	49.9	55.5	60.8	63.6	65.4	66.5	63.6	60.4	56.1	53.3	(20)	
	2%	DB	51.1	50.7	53.6	60.1	65.1	70.1	73.2	71.7	67.8	62.7	57.3	53.8	(21)	
		MCWB	49.6	48.2	48.6	53.0	58.0	61.2	64.3	64.3	62.0	59.0	55.2	52.2	(22)	
	5%	DB	50.1	49.4	51.5	57.1	62.0	67.0	70.6	69.1	65.8	61.3	56.3	52.8	(23)	
		MCWB	48.7	47.4	47.7	51.5	56.4	60.2	63.2	62.8	60.7	58.2	54.3	51.3	(24)	
	10%	DB	49.3	48.4	50.1	54.7	59.6	64.6	68.1	67.1	64.3	60.2	55.3	51.6	(25)	
		MCWB	47.9	46.6	47.2	50.2	54.8	58.9	62.1	61.8	59.9	57.4	53.3	50.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	50.4	51.2	56.9	62.3	65.6	67.1	68.2	64.8	61.7	57.4	53.9	(27)	
		MCDB	51.5	50.8	54.7	60.8	67.1	72.0	73.9	73.5	68.1	63.8	57.8	54.4	(28)	
	2%	WB	50.3	49.3	49.9	54.2	59.3	62.7	65.3	65.4	63.3	60.3	56.0	52.9	(29)	
		MCDB	50.7	49.9	52.0	58.7	63.2	67.8	70.6	69.4	66.0	61.7	56.7	53.5	(30)	
	5%	WB	49.3	48.1	48.9	52.3	57.4	61.1	64.1	64.1	62.2	59.2	54.9	51.8	(31)	
		MCDB	49.9	48.8	50.6	56.0	60.7	65.6	68.6	67.3	64.6	60.7	55.9	52.5	(32)	
	10%	WB	48.4	47.1	48.0	50.9	55.7	59.8	63.0	63.0	61.1	58.1	53.7	50.4	(33)	
		MCDB	49.0	48.1	49.5	53.8	58.7	63.5	66.8	65.8	63.5	59.8	55.0	51.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.5	7.5	8.7	8.8	8.9	8.7	8.5	8.5	7.2	6.6	6.1	(35)	
		MCDBR	5.9	6.5	8.8	11.4	12.1	12.6	12.4	11.3	10.2	7.2	6.2	6.2	(36)	
	5% WB	MCWBR	6.5	5.7	6.4	7.2	7.5	7.4	7.0	6.9	6.8	5.8	6.0	6.6	(37)	
		MCWBR	5.9	5.8	7.4	9.8	10.6	10.6	10.2	9.2	8.2	6.2	5.8	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.331	0.343	0.372	0.388	0.395	0.385	0.386	0.371	0.364	0.353	0.329	0.320	(40)	
		taud	2.350	2.389	2.323	2.305	2.327	2.386	2.387	2.445	2.454	2.448	2.407	2.325	(41)	
	Ebn at Noon	Ebn at Noon	213	246	259	269	272	276	273	272	261	239	213	194	(42)	
		Edh at Noon	20	25	32	37	38	36	36	32	28	24	19	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	242	475	851	1335	1638	1780	1645	1367	1021	572	296	186	(44)	
	RadStd	RadStd	23	50	92	123	93	169	146	105	76	48	26	23	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page